

Sundar Gurumurthy

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Professional Summary

Simulation engineer specialising in **first-principles, physics-based modelling** and **scientific software development** for complex manufacturing and mechanical systems. Experienced in deriving and implementing **thermal, mechanical, and phase-transformation models governed by PDEs**, and translating them into **robust, reduced-order simulation tools** for engineering decision-making. Strong background in **Python/C++ development, numerical methods, multi-objective optimisation, and statistical analysis**, with hands-on experience bridging **theory, experiments, and production-grade software**.

Core Competencies

Simulation & Modelling

- Physics-based modelling (thermal, mechanical, non-linear, contact)
- PDE-based modelling; sparse numerical solvers (SuiteSparse); Monte Carlo methods
- Governing equations and first-principles model formulation
- Reduced-order and surrogate modelling

Programming & Data Science

- Numerical optimisation, inverse problems, regression, uncertainty analysis
- Structured scientific data pipelines
- Automating decision-making through physics- and data-driven models.

Technical Skills

Languages and Frameworks

- **Languages:** Python, C++, Bash/Linux, Rust, Fortran
- **Data Science/ML:** Scikit-learn, PyTorch3D, Pandas, Polars
- **Visualisation:** Plotly, Matplotlib, Seaborn, PyVista, VisPy, GNUplot

Other Tools

- **DevOps:** GitLab CI/CD, GitHub Actions, Docker, Apptainer, PBS (HPC Scheduler)
- **Databases:** MariaDB, HDF5
- **Version Control:** Git, UV, Poetry, cargo (Rust)
- **CAD/PLM:** Siemens NX, CATIA V5, Teamcenter
- **FEA Solvers:** Abaqus, LS-Dyna, Ansys, Nastran

Projects & Applied Modelling

Geometry-Linked Manufacturing Acceptance Criteria for Stress Performance

Cranfield University

2025 – Present

- Developed **geometry-based acceptance limits** linking surface profile metrics to tensile response and distortion behaviour.
- Integrated **physics-based simulation, laser scanning, and mechanical testing** into lightweight validation workflows.
- Replaced repeated high-fidelity simulations with a **fast, model-informed decision framework**.

Python Library for WAAM & Scan-Based Qualification

Cranfield University

2025 – Present

- Designed a modular Python package for **laser-scan processing, geometric feature extraction, and statistical evaluation**.
- Implemented **MVEE-based process window estimation** and deviation mapping for qualification workflows.
- Automated end-to-end pipelines from **scan acquisition → analysis → reporting**.
- Delivered **interactive 2D/3D dashboards** used by industrial partners for process qualification decisions.

Process-Driven WAAM Simulation & Optimisation

Cranfield University

2024 – Present

- Built **thermal–mechanical simulation models** to study process sensitivities and parameter interactions.
- Designed **DOE-based studies** to quantify process–response relationships.
- Provided modelling insight for aerospace and energy components under development.

Work Experience

Research Assistant

Cranfield University, UK

Jun 2024 – Present

- Developed **Python/C++ simulation automation pipelines** for thermal–mechanical analysis and sensitivity studies.
- Formulated **phase-transformation strain models** using test-driven fitting and thermal expansion datasets.
- Integrated **laser-scan geometry** into simulation workflows for deviation-driven modelling.
- Implemented **rapid reduced-order approaches** delivering engineering results in minutes.
- Developed **FORTTRAN heat-source models** and **C++ Abaqus subroutines** for non-standard physics.
- Communicated results via **technical reports, validation studies**, and multidisciplinary reviews.

Graduate Engineering Trainee

Sona Comstar, India

Jul 2021 – Jul 2022

- Developed **custom simulation software** for mechanical contact analysis of complex gear geometries.
- Implemented **continuum-mechanics–based contact and deformation models**, solving governing equations for elastic material response under load.
- Modelled **load-dependent surface contact, stress distribution, and transmission behaviour** using discretised contact formulations.
- Integrated analysis workflows with **Siemens NX** for parametric geometry generation and **Teamcenter** for versioned data management.

Student Trainee – Crash Structures

Mercedes-Benz R&D India

Feb 2021 – Jun 2021

- Built **LS-Dyna models** for tyre, rim, and composite crash scenarios.
- Optimised **Lagrangian contact formulations** for fibre-reinforced rubber composites.
- Automated **ANSA meshing workflows** using Python.
- Supported **test–simulation correlation** for impact and modal behaviour.

Education

MSc by Research, Manufacturing

Cranfield University

Jan 2023 – Oct 2024

Thesis: Improving the Inherent Strain Method for WAAM Simulation

- Reduced distortion prediction error **14% → 3%** through algorithmic model improvements.
- Developed **fast inherent-strain workflows** reducing simulation runtime from days to minutes.

B.E. Mechanical Engineering

BITS Pilani

Aug 2017 – Jun 2021

- CGPA: 7.71/10 (First Class)

Awards

- AIAA/USU SmallSat Travel Award (Blue Origin Sponsored)

References

Available on request.